Dorey's Resource Book

Ratios and Rates

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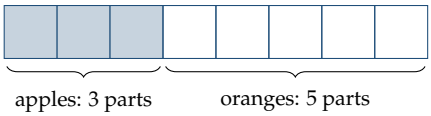
1.01 Ratios and fractions

SYLLABUS MST-12-S1-05

- Recognise ratios and explain their relationship with fractions

NOTES

- A **ratio** compares two or more quantities. It is written $a : b$ and read "a to b".
- Order matters: 3 apples and 5 oranges gives apples to oranges $= 3 : 5$, but oranges to apples $= 5 : 3$.
- Total parts $=$ the sum of the numbers in the ratio: $3 : 5$ has $3 + 5 = 8$ parts.
- Fraction of the total $= \dfrac{\text{part}}{\text{total parts}}$: apples are $\dfrac{3}{8}$ of the fruit.
- In $a : b$, the first quantity is $\dfrac{a}{b}$ of the second: there are $\dfrac{3}{5}$ as many apples as oranges.



WORKED EXAMPLE

A pet shop has 4 cats and 7 dogs. Write the ratio of cats to dogs.

cats : dogs $= 4 : 7$ — cats first, because cats are named first

Tier 1 · Write a ratio

1. An excursion has 5 teachers and 23 students. Write the ratio of teachers to students.

2. A soccer team had 6 wins and 1 loss. Write the ratio of wins to losses.

3. A recipe uses 3 cups of flour and 2 cups of sugar. Write the ratio of flour to sugar.

Tier 2 · Order matters

4. A class has 7 boys and 9 girls. Write the ratio of girls to boys.

5. A car yard has 4 red cars and 11 white cars. Write the ratio of white cars to red cars.

6. A pancake recipe uses 2 eggs and 5 cups of milk. Write the ratio of cups of milk to eggs.

Tier 3 · Three-part ratios

7. A family has 2 cats, 3 dogs and 1 rabbit. Write the ratio of cats to dogs to rabbits.

8. A concrete mix uses 1 bucket of cement, 2 buckets of sand and 3 buckets of gravel. Write the ratio of cement to sand to gravel.

9. A school won 5 gold, 7 silver and 4 bronze medals. Write the ratio of gold to bronze to silver medals.

Tier 4 · Fraction of the total

10. A pencil case has 3 red pens and 4 blue pens. What fraction of the pens are red?

11. In one week there were 5 sunny days and 2 rainy days. What fraction of the days were rainy?

12. A party has 9 adults and 4 children. What fraction of the people are children?

Tier 5 · From a ratio only

13. At an animal shelter, the ratio of cats to dogs is 2 : 5. What fraction of the animals are dogs?

14. A mortar mix has sand and cement in the ratio 4 : 1. What fraction of the mix is cement?

15. A basketball team's ratio of wins to losses is 7 : 2. What fraction of its games did the team win?

Tier 6 · One part as a fraction of the other

16. The ratio of boys to girls in a choir is 3 : 8. The number of boys is what fraction of the number of girls?

17. A drink has cordial and water in the ratio 1 : 5. The amount of cordial is what fraction of the amount of water?

18. Ella's savings and spending are in the ratio 2 : 9. Her savings are what fraction of her spending?

Tier 7 · From a fraction to a ratio

19. $\frac{3}{8}$ of the students catch a bus to school. The rest do not. Write the ratio of bus students to non-bus students.

20. $\frac{1}{4}$ of a backyard is lawn. The rest is garden beds. Write the ratio of lawn to garden beds.

21. $\frac{5}{9}$ of a rugby team are forwards. The rest are backs. Write the ratio of forwards to backs.

Conceptual

22. A fruit bowl has apples and oranges in the ratio 3 : 5. Jo says $\frac{3}{5}$ of the fruit is apples. Explain Jo's mistake.

23. A class has boys and girls in the ratio 2 : 7. Is this the same as the ratio 7 : 2? Explain.

24. In a class, $\frac{5}{12}$ of the students are girls. Write the ratio of girls to boys.

25. A netball team's ratio of wins to losses is 5 : 3. There were no draws.

(a) What fraction of its games did the team lose?

(b) Did the team win more than half of its games? Explain.

26. A café sells tea and coffee in the ratio 2 : 5.

(a) Write the ratio of coffee to tea. 1 mark

(b) What fraction of the drinks sold are tea? 1 mark

(c) Sam says "tea is $\frac{2}{5}$ of the drinks sold". Explain why Sam is wrong. 2 marks

Revision

27. Find 10% of \$340.

28. Find 25% of \$86.

29. Increase \$60 by 15%.

30. Decrease \$250 by 20%.

31. Kai earns \$24.50 per hour. How much does he earn for 18 hours of work?

32. Mia's normal rate of pay is \$22 per hour. How much does she earn for 4 hours at time-and-a-half?

33. Convert 3.5 m to centimetres.

34. Convert 2400 g to kilograms.

35. Convert 750 mL to litres.

36. Convert 1.2 km to metres.

37. The cost of a taxi is $C = 5n + 20$, where n is the number of kilometres. Find C when $n = 8$.

38. Use $A = lw$ to find A when $l = 12$ and $w = 7.5$.

39. Solve $x + 9 = 23$.

40. Solve $4x = 36$.

41. Find the perimeter of a rectangle 8 m long and 5 m wide.

42. Find the area of a rectangle 8 m long and 5 m wide.

43. A shift starts at 9:45 am and finishes at 2:20 pm. How long is the shift?

44. Write 2.25 hours in hours and minutes.

45. Find the mean of 4, 7, 9, 10, 5.

46. Find the median of 3, 8, 6, 2, 9.

47. Find the range of 12, 5, 19, 8.

48. A standard six-sided die is rolled. What is the probability of rolling a 4?

49. A bag has 3 red and 7 blue counters. One is chosen at random. What is the probability it is red?

50. A jacket costs $45 before GST. GST is 10%. Find the price including GST.

51. A phone plan costs $35 per month. How much does it cost for a year?

52. Round 3.476 to two decimal places.

1.02 Simplifying ratios

SYLLABUS MST-12-S1-05

- Express a ratio in simplest form
- Determine the ratio of two quantities

NOTES

- Multiply or divide **every** part of a ratio by the same number to get an **equivalent ratio**: $6 : 8 = 3 : 4$.
- A ratio is in **simplest form** when no number except 1 divides into every part.
- Fastest method: divide every part by the highest common factor (HCF).
- Quantities must be in the same units first: $50\text{ cm} : 2\text{ m} = 50 : 200 = 1 : 4$.
- Decimals: multiply every part by 10 (or 100) first: $2.4 : 1.6 = 24 : 16 = 3 : 2$.
- Fractions: multiply every part by the common denominator: $\frac{1}{2} : \frac{3}{4} = 2 : 3$ (multiply by 4).
- Calculator check: enter $a : b$ as the fraction $\frac{a}{b}$. The simplified fraction gives the simplified ratio.

WORKED EXAMPLE

Simplify $10 : 14$.

The HCF of 10 and 14 is 2.
$10 : 14 = 5 : 7$

2 divides into both
divide both parts by 2

Tier 1 · One common factor

53. Simplify $4 : 10$.

54. Simplify $9 : 6$.

55. Simplify $15 : 20$.

Tier 2 · Highest common factor

56. Simplify $24 : 36$.

57. Simplify $18 : 45$.

58. Simplify $30 : 48$.

Tier 3 · Three-part ratios

59. Simplify 6 : 9 : 15.

60. Simplify 12 : 16 : 20.

61. Simplify 14 : 7 : 21.

Tier 4 · Ratio of two quantities

62. An animal shelter has 12 cats and 20 dogs. Write the ratio of cats to dogs in simplest form.

63. A concert has 40 adults and 16 children. Write the ratio of adults to children in simplest form.

64. A recipe uses 250 g of butter and 400 g of flour. Write the ratio of butter to flour in simplest form.

Tier 5 · Different units

65. Write the ratio of 30 minutes to 2 hours in simplest form.

66. Write the ratio of 400 g to 1.2 kg in simplest form.

67. Write the ratio of 600 mL to 2 L in simplest form.

Tier 6 · Decimals

68. Simplify 0.6 : 1.5.

69. Simplify 2.5 : 4.

70. Simplify 1.2 : 0.8.

Tier 7 · Fractions

71. Simplify $\frac{1}{3} : \frac{1}{2}$.

72. Simplify $\frac{2}{5} : \frac{3}{10}$.

73. Simplify $\frac{3}{4} : \frac{1}{2}$.

Conceptual

74. Ali simplifies 12 : 16 to 6 : 8 and stops. Is his answer in simplest form? Explain.

75. Are the ratios 4 : 6 and 10 : 15 equivalent? Justify your answer.

76. A ratio simplifies to 2 : 5. The first number in the original ratio is 8. What is the second number?

77. A painter mixes 500 mL of blue paint with 1.5 L of white paint. Write the ratio of blue to white paint in simplest form.

78. A class has 18 girls and 12 boys.

(a) Write the ratio of girls to boys in simplest form. 1 mark

(b) Two more boys join the class. Write the new ratio of girls to boys in simplest form. 2 marks

(c) Kim says 18 : 12 simplifies to 9 : 6. Explain why Kim's answer is not in simplest form. 1 mark

Revision

79. A bag has 5 red and 8 blue marbles. Write the ratio of red to blue marbles.

80. The ratio of cats to dogs is 3 : 4. What fraction of the animals are cats?

81. $\frac{2}{9}$ of the students walk to school. The rest do not. Write the ratio of walkers to non-walkers.

82. How many parts are there in total in the ratio 4 : 1 : 3?

83. Find 15% of \$80.

84. Find 30% of \$250.

85. Increase \$120 by 10%.

86. Decrease \$90 by 25%.

87. Write 18 out of 24 as a percentage.

88. Sam earns \$26 per hour. How much does he earn for 35 hours?

89. Lee's normal rate is \$30 per hour. How much does she earn for 3 hours at double time?

90. An annual salary is \$62 400. Find the weekly pay (52 weeks in a year).

91. Convert 4.2 kg to grams.

92. Convert 350 cm to metres.

93. Convert 2.5 L to millilitres.

94. Convert 90 minutes to hours.

95. Use $P = 2l + 2w$ to find P when $l = 9$ and $w = 4$.

96. A plumber charges $C = 40 + 15h$ dollars for h hours. Find C when $h = 3$.

97. Solve $x - 7 = 12$.

98. Solve $\frac{x}{5} = 6$.

99. Find the area of a rectangle 12 cm long and 4.5 cm wide.

100. Find the mean of $12,\ 15,\ 9,\ 14$.

101. Find the mode of $3,\ 5,\ 5,\ 7,\ 8,\ 5,\ 3$.

102. Find the median of $4,\ 9,\ 2,\ 7$.

103. A spinner has 8 equal sections. 3 are red. What is the probability of not landing on red?

104. A bus leaves at 10:50 am and arrives at 1:15 pm. How long is the trip?

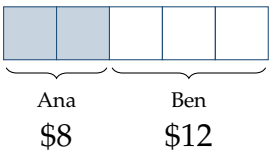
1.03 Dividing in a given ratio

SYLLABUS MST-12-S1-05

- Divide a quantity in a given ratio
- Solve practical problems involving ratios

NOTES

- To divide a quantity in a ratio: find the total number of parts, find the value of one part, then multiply each number in the ratio by one part.
- One part $= \dfrac{\text{total}}{\text{total parts}}$: \$20 in the ratio 2 : 3 is 5 parts, so one part = \$4.
- Check: the shares must match the ratio **and** add up to the total (\$8 + \$12 = \$20).
- Given one share instead of the total: divide that share by its number in the ratio to find one part.
- Given the difference between two shares: divide the difference by the difference in the ratio numbers to find one part.



WORKED EXAMPLE

Share \$45 *between Lily and Max in the ratio* 4 : 5.

Total parts $= 4 + 5 = 9$ — add the numbers in the ratio
One part $= \$45 \div 9 = \$5$ — divide the total by the parts
Lily $= 4 \times \$5 = \$20$, Max $= 5 \times \$5 = \$25$ — multiply each number by one part
Check: $\$20 + \$25 = \$45$ — shares add to the total

Tier 1 · Share in a two-part ratio

105. Share \$30 between Zoe and Liam in the ratio 2 : 3.

106. Share \$48 between Ava and Noah in the ratio 1 : 5.

107. Share 56 lollies between two friends in the ratio 3 : 4.

Tier 2 · Larger quantities

108. Share \$132 in the ratio 7 : 4.

109. Share \$360 in the ratio 5 : 3.

110. Divide 625 g of flour in the ratio 3 : 2.

Tier 3 · Three-part ratios

111. Share \$120 in the ratio 1 : 2 : 3.

112. Divide 180 L of water between three tanks in the ratio 2 : 3 : 4.

113. Share \$500 between Ali, Bec and Cam in the ratio 3 : 5 : 2.

Tier 4 · Given one share

114. The ratio of cats to dogs at a shelter is 2 : 5. There are 14 cats. How many dogs are there?

115. Mortar is mixed with sand and cement in the ratio 4 : 1. A builder uses 3 bags of cement. How many bags of sand are needed?

116. The ratio of adults to children at a show is 5 : 3. There are 24 children. How many adults are there?

Tier 5 · Practical problems with units

117. Cement and sand are mixed in the ratio 1 : 4. How much of each is needed to make 2.5 kg of mix?

118. A lawnmower uses oil and petrol in the ratio 1 : 25. How many millilitres of oil are in 2.6 L of fuel mix?

119. Red and white paint are mixed in the ratio 2 : 3 to make 4 L of pink paint. How much of each colour is used?

Tier 6 · Find the difference

120. $600 is shared between Jai and Ruby in the ratio 1 : 3. How much more does Ruby get than Jai?

121. A school group of 72 students has girls and boys in the ratio 5 : 4. How many more girls than boys are there?

122. Two business partners share a $2400 profit in the ratio 3 : 5. What is the difference between their shares?

Tier 7 · Work back from the difference

123. Two shares are in the ratio 2 : 5. The difference between them is $60. What is the total amount?

124. Two bags of rice have masses in the ratio 4 : 7. The heavier bag is 15 kg heavier. What is the total mass?

125. Mel and Dan share some money in the ratio 1 : 4. Dan gets $36 more than Mel. How much does each person get?

Conceptual

126. Ella shares $40 in the ratio 3 : 5 and writes the shares as $3 and $5. Explain her mistake and find the correct shares.

127. Sam shares $50 between Pia and Rob in the ratio 2 : 3. He gives Pia $30 and Rob $20. Is he correct? Explain.

128. Can $50 be shared exactly in the ratio 3 : 4 using whole dollars? Explain.

129. A grandparent shares $9000 between three grandchildren in the ratio of their ages: 10, 12 and 14. How much does the eldest grandchild receive?

130. A mortar mix uses sand and cement in the ratio 5 : 1.

(a) What fraction of the mix is cement? 1 mark

(b) How much sand is in 48 kg of mix? 2 marks

(c) A builder has 15 kg of cement. How much sand is needed? 1 mark

Revision

131. A fruit stall has 7 apples and 2 pears. Write the ratio of apples to pears.

132. The ratio of tea to coffee sold is 2 : 9. What fraction of the drinks sold are coffee?

133. $\frac{3}{10}$ of the staff walk to work. The rest do not. Write the ratio of walkers to non-walkers.

134. Simplify 18 : 24.

135. Simplify 35 : 21.

136. Simplify 8 : 12 : 20.

137. Write the ratio of 45 minutes to 2 hours in simplest form.

138. Simplify 0.4 : 1.8.

139. Simplify $\frac{1}{4} : \frac{2}{3}$.

140. Find 20% of \$65.

141. Increase \$80 by 5%.

142. Decrease \$150 by 40%.

143. Priya earns \$28.40 per hour. How much does she earn for 7.5 hours?

144. Ben's normal rate is \$24 per hour. How much does he earn for 6 hours at time-and-a-half?

145. A TV costs \$250 before GST. GST is 10%. Find the price including GST.

146. Convert 3.75 km to metres.

147. Convert 0.85 L to millilitres.

148. Convert 6500 g to kilograms.

149. Use $v = u + at$ to find v when $u = 4$, $a = 3$ and $t = 5$.

150. Solve $3x + 4 = 19$.

151. Find the area of a triangle with base 10 cm and height 6 cm.

152. Find the perimeter of a square with side length 7.5 m.

153. Find the mean of $6,\ 8,\ 10,\ 12,\ 14$.

154. Find the range of $23,\ 8,\ 15,\ 31$.

155. A card is drawn at random from a standard deck of 52 cards. What is the probability that it is a heart?

156. Round 12.6849 to one decimal place.

1.04 Map scales

SYLLABUS MST-12-S1-05

- Explain the relationship between ratios and map scales and apply this relationship to solve problems in a variety of practical contexts

NOTES

- A map scale of $1 : n$ means 1 unit on the map represents n of the **same** units in real life: $1 : 1000$ means 1 cm represents 1000 cm $=$ 10 m.
- To compare scales, write both parts in the same unit: $2\text{ cm} : 100\text{ m} = 2 : 10\,000 = 1 : 5000$.
- Conversions: $100\text{ cm} = 1\text{ m}$, $1000\text{ m} = 1\text{ km}$, $100\,000\text{ cm} = 1\text{ km}$.
- Map to real life: **multiply** by n. Real life to map: **divide** by n (in the same unit).
- A scale bar shows what one length on the map represents. Count how many of those lengths fit, then multiply.

WORKED EXAMPLE

A map has a scale of $1 : 200$. *A distance on the map is 7 cm. What is the actual distance in centimetres?*

$\text{Actual} = 7 \times 200$ — map to real life: multiply by n
$= 1400\text{ cm}$ — same unit as the map

Tier 1 · Map to real life

157. A plan has a scale of $1 : 50$. A distance on the plan is 9 cm. What is the actual distance in centimetres?

158. A map has a scale of $1 : 300$. A distance on the map is 2.5 cm. What is the actual distance in centimetres?

159. A map has a scale of $1 : 400$. A distance on the map is 6 cm. What is the actual distance in centimetres?

Tier 2 · Answer in metres

160. A map has a scale of 1 : 1000. A distance on the map is 8 cm. What is the actual distance in metres?

161. A park map has a scale of 1 : 500. A path on the map is 4.2 cm long. How long is the path in metres?

162. A map has a scale of 1 : 2000. Two buildings are 3 cm apart on the map. How far apart are they in metres?

Tier 3 · Answer in kilometres

163. A map has a scale of 1 : 100 000. Two towns are 6 cm apart on the map. How far apart are they in km?

164. A bushwalking map has a scale of 1 : 25 000. A track is 8.4 cm long on the map. How long is the track in km?

165. A road map has a scale of 1 : 250 000. A road is 3.2 cm long on the map. How long is the road in km?

Tier 4 · Real life to map

166. A plan has a scale of 1 : 100. A room is 9 m long. How long is the room on the plan?

167. A map has a scale of 1 : 50 000. Two towns are 4 km apart. How far apart are they on the map?

168. A map has a scale of 1 : 20 000. A beach is 1.5 km long. How long is the beach on the map?

Tier 5 · Find the scale

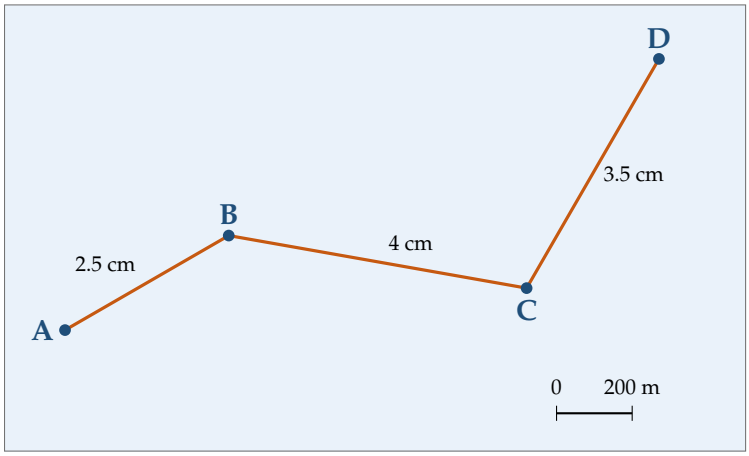
169. On a plan, 2 cm represents 10 m. Write the scale in the form 1 : n.

170. On a map, 5 cm represents 1 km. Write the scale in the form 1 : n.

171. On a map, 3 cm represents 15 km. Write the scale in the form 1 : n.

Tier 6 · Scale bars and routes

Use the map for the next three questions. The scale bar shows that 1 cm represents 200 m.



172. How far is it from A to B in real life?

173. How far is the route A to B to C in real life? Give your answer in km.

174. How long is the whole route from A to D in real life?

Conceptual

175. Ari says a scale of 1 : 1000 means 1 cm on the map represents 1000 m. Explain his mistake.

176. Which map shows more detail: a map with scale 1 : 10 000 or a map with scale 1 : 1 000 000? Explain.

177. Two towns are 12 km apart. How far apart are they on a map with scale 1 : 200 000?

178. A school map has a scale of 1 : 500. The oval is 18 cm long on the map. Is the oval long enough for a 100 m straight sprint? Explain.

179. A bushwalking map has a scale of 1 : 25 000.

(a) A track is 6 cm long on the map. How long is it in km? 1 mark

(b) A walk is 4 km long. How long is it on the map? 1 mark

(c) A different map shows the same 4 km walk as 8 cm. Find the scale of this map in the form 1 : n. 2 marks

Revision

180. Simplify 24 : 30.

181. Write the ratio of 5 cm to 2 m in simplest form.

182. Simplify 1.5 : 4.5.

183. Share \$60 in the ratio 1 : 5.

184. Share \$360 in the ratio 4 : 5.

185. Divide 72 kg in the ratio 1 : 2 : 3.

186. The ratio of boys to girls is 2 : 3. There are 18 girls. How many boys are there?

187. The ratio of cats to dogs is 4 : 9. What fraction of the animals are dogs?

188. $\frac{2}{7}$ of a class walk to school. The rest do not. Write the ratio of walkers to non-walkers.

189. Find 35% of \$240.

190. Increase \$45 by 20%.

191. Decrease \$96 by 12.5%.

192. Jess earns \$31.20 per hour. How much does she earn for 12 hours?

193. Omar's normal rate is \$25 per hour. How much does he earn for 4 hours at double time?

194. Convert 0.65 km to metres.

195. Convert 4.8 m to centimetres.

196. Convert 2500 mL to litres.

197. Solve $2x - 5 = 13$.

198. A hire company charges $C = 25 + 0.4d$ dollars for d km. Find C when $d = 150$.

199. Find the area of a rectangle 6.5 m long and 4 m wide.

200. Find the mean of 15, 18, 21, 26.

201. Find the median of 7, 3, 9, 12, 5, 8.

202. A standard six-sided die is rolled. What is the probability of rolling an even number?

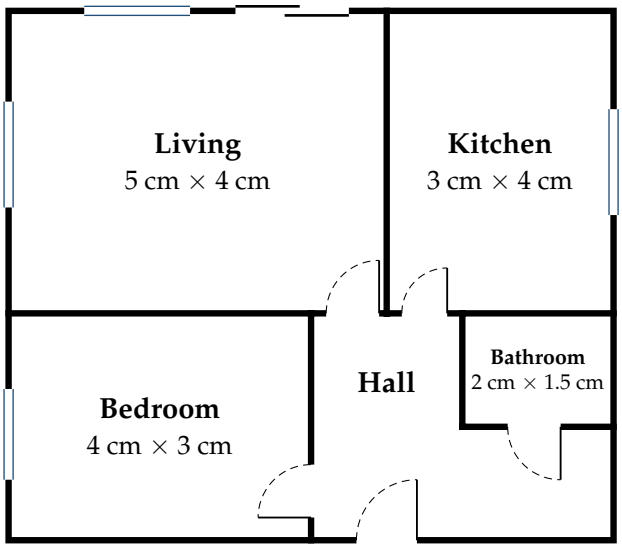
1.05 Scale drawings and building plans

SYLLABUS MST-12-S1-05

- Obtain measurements from scale drawings and building plans with commonly used symbols to solve problems in a variety of practical contexts
- Estimate and compare quantities, materials and costs using measurements taken from scale drawings

NOTES

- Building plans use symbols: a thick line is a wall, a double thin line is a window, a gap with a quarter-circle is a door, and two overlapping lines in a gap is a sliding door.
- A 1 : 100 plan means 1 cm on the plan represents 100 cm = 1 m. A 1 : 50 plan means 1 cm represents 50 cm.
- Change plan lengths to real lengths **first**, then find areas: real area = real length × real width.
- Cost = area × price per m^2. Add any fixed charges, such as delivery.
- You can't buy part of a tin, box or roll, so always round **up** when buying materials.



Scale 1 : 100

WORKED EXAMPLE

A plan of a laundry has a scale of 1 : 100. *On the plan the laundry is* 2.5 *cm long and 2 cm wide. Find its real length and width in metres.*

Length = 2.5 × 100 = 250 cm = 2.5 m — plan to real: multiply by 100
Width = 2 × 100 = 200 cm = 2 m — change cm to m

Tier 1 · Real lengths from a plan

203. Use the floor plan above. Find the real length and width of the kitchen in metres.

204. A plan has a scale of 1 : 100. A garage is 6 cm by 3.5 cm on the plan. Find its real length and width.

205. A furniture plan has a scale of 1 : 50. A desk is 3 cm by 1.6 cm on the plan. Find the real length and width of the desk in metres.

Tier 2 · Real area

206. Find the real floor area of the kitchen.

207. A plan has a scale of 1 : 100. A garage is 6 cm by 3.5 cm on the plan. Find its real floor area.

208. A plan has a scale of 1 : 100. A patio is 4.5 cm by 3 cm on the plan. Find its real area.

Tier 3 · Cost from area

209. Floor tiles cost \$28 per m^2. Find the cost to tile the kitchen.

210. Sealing a concrete floor costs \$18 per m^2. Find the cost to seal the 21 m^2 garage floor.

211. Pavers cost \$55 per m^2. Find the cost to pave the 13.5 m^2 patio.

Tier 4 · Reading plan symbols

212. Use the floor plan. How many windows does the house have?

213. Which room has a sliding door?

214. How many doors open onto the hall, including the front door?

Tier 5 · Two or more rooms

215. Carpet costs \$40 per m^2. Find the cost to carpet the bedroom and the living room.

216. Tiles cost \$35 per m^2. Find the cost to tile the kitchen and the bathroom.

217. Find the total floor area of the whole house.

Tier 6 · Estimating materials

218. The bathroom (3 m^2) is tiled with square tiles 30 cm by 30 cm. How many tiles are needed?

219. The bedroom walls have a total length of 14 m and are 2.4 m high. One litre of paint covers 16 m^2. How many litres are needed for one coat? (Ignore the door and window.)

220. A backyard plan has a scale of 1 : 200. The lawn is 6 cm by 4 cm on the plan. Each roll of turf covers 1.5 m^2. How many rolls are needed?

Tier 7 · Comparing costs

221. For the 12 m^2 bedroom, which is cheaper: carpet at \$52 per m^2, or vinyl at \$39 per m^2 plus \$120 to install? By how much?

222. The 12 m^2 kitchen is to be tiled. Shop A charges \$30 per m^2. Shop B sells boxes that cover 1.5 m^2 for \$42 a box. Which shop is cheaper, and by how much?

223. A job needs 6 L of paint. A 4 L tin costs \$68 and a 1 L tin costs \$22. Is it cheaper to buy two 4 L tins or six 1 L tins?

Conceptual

224. On a 1 : 100 plan, a room is 4 cm by 3 cm. Mia says the area on the plan is 12 cm^2, so the real area is $12 \times 100 = 1200$ cm^2. Explain her mistake and find the real area.

225. A calculation shows 33.3 tiles are needed. Explain why the answer is not 33 tiles.

226. A plan has a scale of 1 : 50. A room is 8 cm by 6 cm on the plan. Find the real area of the room.

227. A tiler adds 10% extra tiles to allow for breakages. A floor needs 12 tiles. How many tiles should be bought?

228. A plan of a deck has a scale of 1 : 100. The deck is 6 cm by 4 cm on the plan.

(a) Find the real length and width of the deck. 1 mark

(b) Find the real area of the deck. 1 mark

(c) Decking boards cost \$65 per m^2. A tin of stain costs \$45 and covers 10 m^2. Find the total cost of the boards and stain. 2 marks

Revision

229. Simplify 45 : 60.

230. Write the ratio of 80 cm to 2 m in simplest form.

231. Share \$96 in the ratio 5 : 3.

232. Divide 1.5 L of juice in the ratio 1 : 2.

233. A map has a scale of 1 : 25 000. Two points are 4 cm apart on the map. How far apart are they in km?

234. Two towns are 30 km apart. How far apart are they on a map with scale 1 : 200 000?

235. On a map, 2 cm represents 5 km. Write the scale in the form 1 : n.

236. $\frac{3}{5}$ of the cars in a car park are white. Write the ratio of white cars to other cars.

237. The ratio of tea to coffee is 4 : 7. What fraction of the drinks are tea?

238. Find 15% of \$360.

239. Increase \$250 by 8%.

240. A price including 10% GST is \$132. Find the price before GST.

241. Liv works 38 hours a week at \$29.50 per hour. Find her weekly pay.

242. Ned's normal rate is \$32 per hour. Find his pay for 3 hours at time-and-a-half.

243. Convert 1250 mm to metres.

244. Convert 0.8 m to millimetres.

245. Convert 2.75 L to millilitres.

246. Find the perimeter of a rectangle 7.2 m long and 3.5 m wide.

247. Find the area of a triangle with base 8 m and height 5 m.

248. Find the area of a circle with radius 3 m, correct to one decimal place.

249. Solve $5x + 3 = 38$.

250. Use $d = 60t$ to find d when $t = 2.5$.

251. Find the mean of $3,\ 7,\ 8,\ 10,\ 12$.

252. Find the mode of $2,\ 4,\ 4,\ 5,\ 6,\ 6,\ 6$.

253. A bag has 4 red, 6 blue and 2 green counters. One is chosen at random. What is the probability it is red?

254. Round 0.0467 to two decimal places.

1.06 Rates and unit rates

SYLLABUS MST-12-S1-05

- Explain the difference between ratios and rates
- Represent given information as a unit rate

NOTES

- A **ratio** compares quantities measured in the **same** units: 3 cups of flour to 5 cups of sugar is 3 : 5.
- A **rate** compares quantities measured in **different** units, written with "per" or "/": 60 km/h, \$24 per hour.
- A **unit rate** is the amount for **1** of the second quantity. Divide by the second quantity: 180 km in 2 h $= 180 \div 2 = 90$ km/h.
- The units of a rate come from the two quantities: dollars and hours give \$/h, litres and minutes give L/min.
- Write times in decimal hours before dividing: 3 h 30 min $= 3.5$ h, 2 h 15 min $= 2.25$ h.

WORKED EXAMPLE

A cyclist rides 60 km in 3 hours. Write this as a unit rate in km/h.

Unit rate $= 60 \div 3$ — divide by the second quantity
$= 20$ km/h — km per 1 hour

Tier 1 · Distance, volume and time

255. A bus travels 240 km in 4 hours. Write this as a unit rate in km/h.

256. A runner covers 1500 m in 5 minutes. Write this as a unit rate in m/min.

257. A hose delivers 45 L of water in 9 minutes. Write this as a unit rate in L/min.

Tier 2 · Money rates

258. Ruby earns \$217 for 7 hours of work. Find her rate of pay per hour.

259. 12 m of fabric costs \$84. Find the cost per metre.

260. A tradie charges \$1350 for 5 days of work. Find the rate per day.

Tier 3 · Decimal answers

261. A 4 kg bag of potatoes costs \$13.80. Find the cost per kilogram.

262. 6 L of milk costs \$10.50. Find the cost per litre.

263. 8 m of garden edging costs \$27. Find the cost per metre, to the nearest cent.

Tier 4 · Rates per minute or per hour

264. A pulse is measured as 18 beats in 15 seconds. Write this as a rate in beats per minute.

265. A drip fills 250 mL in 30 seconds. Write this as a rate in mL per minute.

266. A cyclist rides 20 km in 15 minutes. Write this as a rate in km/h.

Tier 5 · Dividing by a decimal

267. A 2.5 kg leg of lamb costs \$21.25. Find the cost per kilogram.

268. A 1.5 L bottle of juice costs \$4.20. Find the cost per litre.

269. 0.4 kg of cheese costs \$6. Find the cost per kilogram.

Tier 6 · Hours and minutes

270. A car travels 315 km in 3 hours 30 minutes. Find its average speed in km/h.

271. Josh earns \$57 for 2 hours 30 minutes of work. Find his rate of pay per hour.

272. A truck travels 220 km in 2 hours 45 minutes. Find its average speed in km/h.

Conceptual

273. Explain the difference between a ratio and a rate. Give an example of each.

274. A recipe uses 3 cups of rice to 5 cups of water. Is this a ratio or a rate? Explain.

275. Jai says a car that travels 200 km in 4 hours has a speed of $4 \div 200 = 0.02$ km/h. Explain his mistake and find the correct speed.

276. Hiring a car costs \$195 for 3 days. Find the cost per day, then the cost to hire the car for 5 days at the same rate.

277. A tank leaks 36 L of water in 4 hours.

(a) Write the leak as a unit rate in L/h. 1 mark

(b) How much water leaks in one day? 1 mark

(c) Water costs \$3.20 per 1000 L. Find the cost of the water lost in one week, to the nearest cent. 2 marks

Revision

278. Simplify 36 : 48.

279. Write the ratio of 250 g to 1 kg in simplest form.

280. Share \$140 in the ratio 3 : 4.

281. The ratio of boys to girls is 3 : 5. There are 15 boys. How many girls are there?

282. A map has a scale of 1 : 50 000. A road is 7 cm long on the map. How long is it in km?

283. A map has a scale of 1 : 20 000. A path is 3 km long. How long is it on the map?

284. On a 1 : 100 plan, a room is 4.5 cm by 3 cm. Find its real area.

285. A job needs 5.2 tins of paint. How many tins must be bought?

286. Carpet costs \$38 per m^2. Find the cost to carpet a 15 m^2 room.

287. The ratio of red to blue counters is 2 : 7. What fraction are red?

288. Find 12.5% of \$480.

289. Increase \$75 by 40%.

290. An annual salary is \$71 500. Find the fortnightly pay (26 fortnights in a year).

291. Sana's normal rate is \$27 per hour. Find her pay for 5 hours at double time.

292. Convert 3.6 km to metres.

293. Convert 425 cm to metres.

294. Write 95 minutes in hours and minutes.

295. Solve $\frac{x}{4} = 9$.

296. Use $A = \frac{1}{2}bh$ to find A when $b = 12$ and $h = 7$.

297. Find the area of a square with side length 2.5 m.

298. Find the mean of $9,\ 11,\ 14,\ 18$.

299. Find the range of $41,\ 17,\ 33,\ 25$.

300. A card is drawn from a standard deck of 52 cards. What is the probability that it is a king?

1.07 Converting rates

SYLLABUS MST-12-S1-05

- Convert between rates

NOTES

- Useful facts: 1 km = 1000 m, 1 L = 1000 mL, 1 h = 60 min = 3600 s, \$1 = 100 c.
- To convert a rate, write it as a fraction and change **one unit at a time**: first the top unit, then the bottom unit.
- Changing the top unit: 2 km/min = 2000 m/min (multiply by 1000).
- Changing the bottom unit: 120 km/h = 120 km per 60 min = 2 km/min (divide by 60).
- km/h to m/s: $\frac{90\,000 \text{ m}}{3600 \text{ s}} = 25$ m/s (divide by 3.6). m/s to km/h: multiply by 3.6.
- To compare two rates, convert them to the **same** units first.

WORKED EXAMPLE

Convert 3 km/min to m/min.

3 km = 3000 m — change the top unit: ×1000
3 km/min = 3000 m/min — the bottom unit stays the same

Tier 1 · Change the top unit

301. Convert 4 km/min to m/min.

302. Convert 2.5 L/h to mL/h.

303. A phone call costs \$3 per minute. Write this rate in cents per minute.

Tier 2 · Change the bottom unit

304. Convert 180 km/h to km/min.

305. Convert 6 m/s to m/min.

306. A pump moves 1200 L/h. Convert this to L/min.

Tier 3 · km/h to m/s

307. Convert 72 km/h to m/s.

308. Convert 36 km/h to m/s.

309. Convert 108 km/h to m/s.

Tier 4 · m/s to km/h

310. Convert 5 m/s to km/h.

311. Convert 25 m/s to km/h.

312. Convert 12 m/s to km/h.

Tier 5 · mL/min to L/h

313. Convert 250 mL/min to L/h.

314. A sprinkler uses 800 mL/min. Convert this to L/h.

315. A leaking pipe loses 75 mL/min. Convert this to L/h.

Tier 6 · L/h to mL/min

316. Convert 6 L/h to mL/min.

317. A drip delivers 1.8 L/h. Convert this to mL/min.

318. A fish tank filter pumps 12 L/h. Convert this to mL/min.

Tier 7 · Compare rates

319. A sprinter runs at 10.4 m/s. An e-scooter travels at 40 km/h. Which is faster?

320. Tap A fills a tank at 10 L/min. Tap B fills it at 540 L/h. Which tap is faster?

321. Drip A delivers 2 mL/min. Drip B delivers 0.15 L/h. Which drip is faster?

Conceptual

322. Explain why you can divide by 3.6 to change km/h to m/s.

323. Ella says $20\ \text{m/s} = 20 \times 3600 \div 1000 = 72\ \text{km/h}$. Is she correct? Explain.

324. The speed limit in a street is 50 km/h. A car is measured at 15 m/s. Is the car speeding? Explain.

325. A hospital drip is set to 125 mL/h. How many mL is that per minute, correct to one decimal place?

326. A garden hose delivers 18 L/min.

(a) Convert this rate to L/h. 1 mark

(b) Convert this rate to mL/s. 1 mark

(c) How long will it take to fill a 2700 L pool? Give your answer in hours and minutes. 2 marks

Revision

327. Is "45 km per hour" a ratio or a rate? Explain.

328. A car travels 210 km in 3 hours. Find the unit rate in km/h.

329. Ava earns \$91 for 7 hours of work. Find her hourly rate.

330. A 2.5 kg bag of rice costs \$18.75. Find the cost per kilogram.

331. A pulse is 24 beats in 20 seconds. Write this in beats per minute.

332. A train travels 260 km in 3 hours 15 minutes. Find its average speed.

333. Simplify 28 : 42.

334. Share \$250 in the ratio 2 : 3.

335. A map has a scale of 1 : 10 000. A road is 8 cm long on the map. How long is it in metres?

336. On a 1 : 100 plan, a deck is 5 cm by 3.2 cm. Find its real area.

337. A job needs 3.4 boxes of tiles. How many boxes must be bought?

338. The ratio of sand to cement is 5 : 3. What fraction of the mix is sand?

339. Find 22% of \$350.

340. Decrease \$64 by 25%.

341. A weekly wage is \$1450. Find the annual wage (52 weeks).

342. Find the pay for 6 hours at time-and-a-half when the normal rate is \$26 per hour.

343. Convert 0.35 L to millilitres.

344. Convert 7.2 km to metres.

345. Convert 150 seconds to minutes.

346. Solve $4x - 7 = 21$.

347. Use $P = 2(l + w)$ to find P when $l = 6$ and $w = 4.5$.

348. Find the volume of a rectangular prism 3 m long, 2 m wide and 1.5 m high.

349. Find the mean of $4,\ 6,\ 6,\ 9,\ 10$.

350. Find the median of $13,\ 8,\ 21,\ 15,\ 10$.

351. A standard die is rolled. What is the probability of not rolling a 6?

352. Round 1234.567 to the nearest whole number.

1.08 Comparing rates and best buys

SYLLABUS MST-12-S1-05

- Make comparisons between rates in a variety of practical contexts
- Compare unit prices to determine best buys

NOTES

- To compare rates, write each one as a **unit rate** in the **same units**, then compare.
- The **unit price** is the price for one unit (per kg, per L, per item, or per 100 g). The **best buy** has the lowest unit price.
- Price per 100 g: divide the price by the number of 100 g in the pack: 500 g for \$4.00 is $\$4.00 \div 5 = \$0.80$ per 100 g.
- Convert first if the sizes are in different units: 750 mL $=$ 0.75 L, 1.5 kg $=$ 1500 g.
- Apply any discount before finding the unit price. The bigger pack is **not always** the better buy.

WORKED EXAMPLE

Which is the better buy: 2 kg of flour for \$4.60*, or 5 kg of flour for* \$10.50*?*

2 kg: $\$4.60 \div 2 = \$2.30/\text{kg}$ — unit price
5 kg: $\$10.50 \div 5 = \$2.10/\text{kg}$ — same units
The 5 kg bag is the better buy. — lower unit price

Tier 1 · Same units

353. Which is the better buy: 2 kg of mince for \$19.80, or 3 kg for \$28.50?

354. Which is the better buy: 4 L of milk for \$6.40, or 2 L for \$3.40?

355. Which is the better buy: 6 toilet rolls for \$4.50, or 9 rolls for \$6.30?

Tier 2 · Different units

356. Which is the better buy: 600 g of grapes for \$4.20, or 1.5 kg for \$9.75?

357. Which is the better buy: 500 mL of oil for \$2.40, or 1.25 L for \$5.50?

358. Which is the better buy: 250 mL of cream for \$1.50, or 1 L for \$5.20?

Tier 3 · Price per 100 g

359. Use the price per 100 g to find the better buy: 150 g of nuts for $2.70, or 400 g for $6.40.

360. Use the price per 100 g to find the better buy: 200 g of ham for $3.60, or 350 g for $6.65.

361. Use the price per 100 g to find the better buy: 500 g of pasta for $8.50, or 1 kg for $16.

Tier 4 · Three options

362. Batteries cost $3.20 for 4, $7.50 for 10, or $16 for 20. Which is the best buy?

363. Detergent costs $2.80 for 1 L, $5.20 for 2 L, or $8.10 for 3 L. Which is the best buy?

364. Tea costs $2.50 for 100 g, $5.75 for 250 g, or $11.75 for 500 g. Which is the best buy?

Tier 5 · Comparing other rates

365. Amir earns $384 for 16 hours. Bella earns $460 for 20 hours. Who has the higher hourly rate?

366. Pump A moves 1200 L in 40 minutes. Pump B moves 1500 L in 1 hour. Which pump is faster?

367. Runner A runs 5 km in 25 minutes. Runner B runs 8 km in 44 minutes. Who is faster?

Tier 6 · Specials and discounts

368. A 1 kg pack of chicken is usually $15 but is 20% off. A 1.5 kg pack costs $17.40. Which is the better buy?

369. A 2 L bottle of juice is usually $6 but is half price. A 3 L bottle costs $4.20. Which is the better buy?

370. A 500 g tub of yoghurt is usually $7 but is 10% off. A 750 g tub costs $9. Which is the better buy?

Tier 7 · Is bigger better?

371. A 2 kg bag of sugar costs \$9.80. A 5 kg bag costs \$25.50. Is the bigger bag the better buy?

372. A 375 mL can of drink costs \$1.80. A 1.25 L bottle costs \$6.25. Which is the better buy?

373. A dozen eggs costs \$6.60. A tray of 30 eggs costs \$17.40. Which is the better buy?

Conceptual

374. Explain why you should compare unit prices rather than the prices on the packs.

375. Jess says the biggest pack is always the best buy. Use an example to show she is wrong.

376. A shelf label shows a unit price of \$1.20 per 100 g for a 450 g pack. What is the price of the pack?

377. Give two reasons someone might choose the pack with the higher unit price.

378. Brand X sells 750 g of rice for \$5.25. Brand Y sells 1.2 kg for \$8.16.

(a) Find the price per 100 g of Brand X. 1 mark

(b) Find the price per 100 g of Brand Y. 1 mark

(c) Kim needs exactly 3 kg of rice and must buy whole packs. Which is the cheapest way to buy at least 3 kg? 2 marks

Revision

379. Convert 54 km/h to m/s.

380. Convert 8 m/s to km/h.

381. Convert 300 mL/min to L/h.

382. Convert 9 L/h to mL/min.

383. A coach travels 420 km in 6 hours. Find the unit rate in km/h.

384. Kate earns \$148.50 for 5.5 hours of work. Find her hourly rate.

385. Is "3 cups of oats to 4 cups of milk" a ratio or a rate?

386. Simplify 1.2 : 3.6.

387. Share 84 marbles in the ratio 3 : 4.

388. A map has a scale of $1 : 25\,000$. A path is 10 cm long on the map. How long is it in km?

389. On a $1 : 100$ plan, a room is 3.5 cm by 4 cm. Find its real area.

390. A job needs 2.1 L of paint, sold in 1 L tins. How many tins must be bought?

391. Find 18% of \$450.

392. Increase \$1200 by 2.5%.

393. Find the GST (10%) on a \$380 item.

394. Ari earns \$1140 for a 38-hour week. Find his hourly rate.

395. Find the pay for 7.5 hours at double time when the normal rate is \$28 per hour.

396. Write 2.5 hours in minutes.

397. Convert 0.45 kg to grams.

398. Solve $3(x + 2) = 21$.

399. Find the circumference of a circle with radius 5 cm, correct to one decimal place.

400. A rectangle has area 36 m^2 and length 9 m. Find its width.

401. Find the mean of $21,\ 25,\ 30,\ 32$.

402. Find the mode of $1,\ 3,\ 3,\ 3,\ 5,\ 7,\ 7$.

403. A coin is tossed. What is the probability of a head?

404. Round 7.995 to two decimal places.

1.09 Speed, distance and time

SYLLABUS MST-12-S1-05

- Use rates to solve problems in a variety of practical contexts
- Interpret and analyse distance–time graphs to solve problems related to speed, distance and time

NOTES

- $\text{Speed} = \dfrac{\text{Distance}}{\text{Time}}$, $\text{Distance} = \text{Speed} \times \text{Time}$, $\text{Time} = \dfrac{\text{Distance}}{\text{Speed}}$.
- Keep the units matching: km with hours gives km/h, m with seconds gives m/s.
- Write minutes as decimal hours before calculating (45 min = 0.75 h). Change a decimal answer back to minutes by multiplying the decimal part by 60 (2.25 h = 2 h 15 min).
- On a distance–time graph, read across from a time or up from a distance. The speed of a straight section is $\dfrac{\text{distance travelled}}{\text{time taken}}$.
- A **steeper** line means a **faster** speed. A **flat** (horizontal) line means the object has **stopped**.

WORKED EXAMPLE

A car travels at 70 km/h for 2 hours. How far does it travel?

$D = S \times T$ — distance formula

$D = 70 \times 2 = 140$ km — speed in km/h, time in hours

Tier 1 · Find the distance

405. A coach travels at 70 km/h for 3 hours. How far does it travel?

406. A cyclist rides at 18 km/h for 2.5 hours. How far does she ride?

407. A walker walks at 5 km/h for 1 hour 30 minutes. How far does he walk?

Tier 2 · Find the time

408. How long does it take to drive 300 km at 75 km/h?

409. How long does it take to ride 36 km at 12 km/h?

410. How long does it take to walk 9 km at 6 km/h?

Tier 3 · Find the speed

411. A car travels 225 km in 2.5 hours. Find its average speed.

412. A rower covers 12 km in 45 minutes. Find the average speed in km/h.

413. A runner covers 400 m in 50 seconds. Find the speed in m/s.

Tier 4 · Hours and minutes

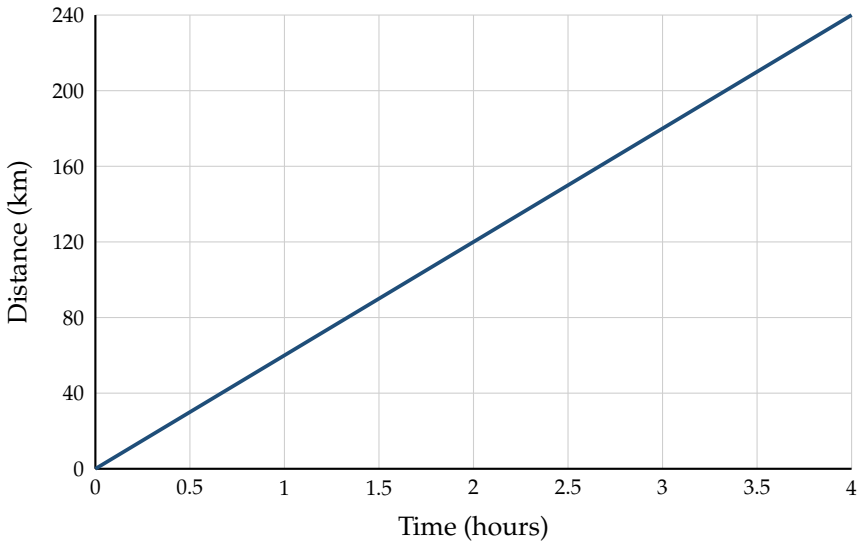
414. How long does it take to travel 100 km at 80 km/h? Give your answer in hours and minutes.

415. How long does it take to travel 50 km at 60 km/h? Give your answer in minutes.

416. How long does it take to travel 245 km at 70 km/h? Give your answer in hours and minutes.

Tier 5 · Reading a distance–time graph

The graph shows a car trip. Use it for the next three questions.



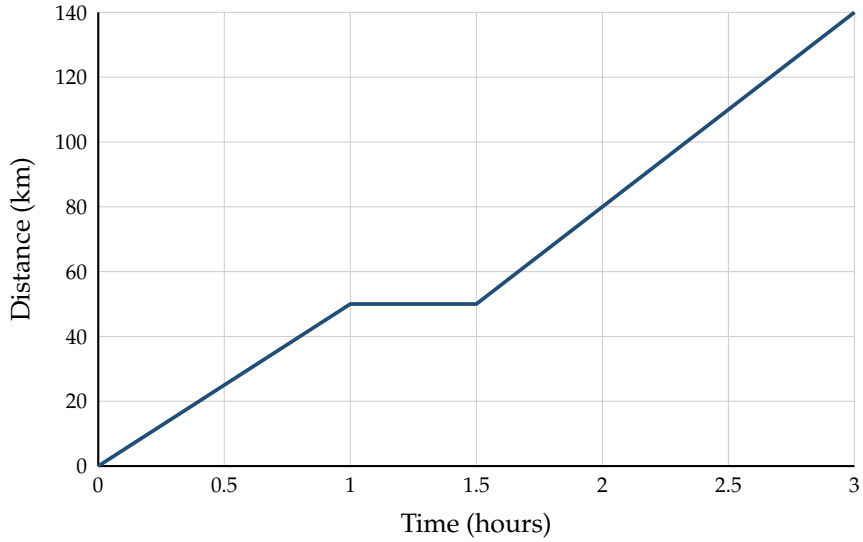
417. How far had the car travelled after 1.5 hours?

418. How long did it take the car to travel 180 km?

419. Find the speed of the car.

Tier 6 · Graphs with a stop

The graph shows a trip with a stop. Use it for the next three questions.



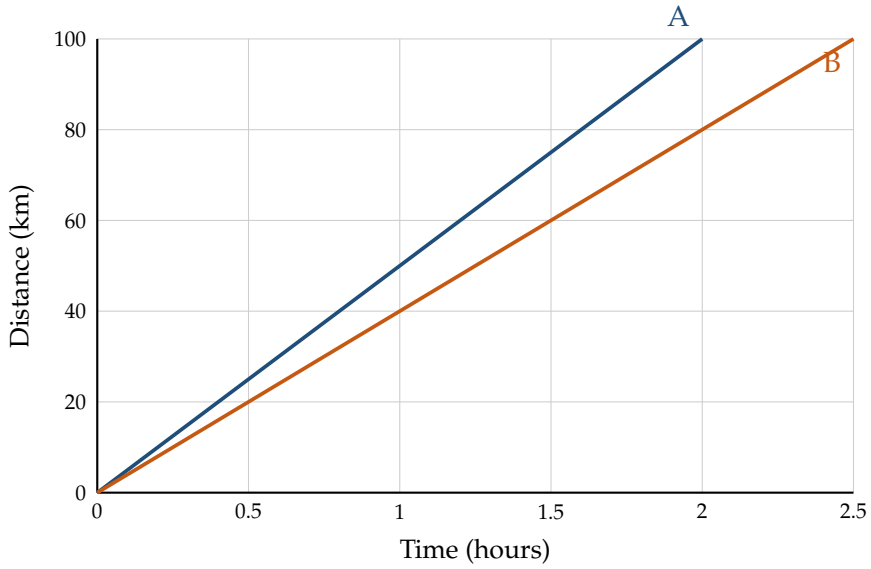
420. Find the speed for the first stage of the trip.

421. How long did the stop last?

422. Find the speed for the last stage of the trip.

Tier 7 · Comparing two travellers

The graph shows two cyclists, A and B, riding 100 km. Use it for the next three questions.



423. Find the speed of each cyclist. Who is faster?

424. How far apart are the cyclists after 2 hours?

425. How much earlier does A finish than B?

Conceptual

426. What does a horizontal (flat) section on a distance–time graph tell you? Explain.

427. Two lines start at the origin on the same distance–time graph. One is steeper. What does this tell you?

428. Sam works out that a 150 km trip at 60 km/h takes 2.5 hours, and writes "2 h 50 min". Explain his mistake.

429. A train leaves at 9:15 am and travels 210 km at 84 km/h. At what time does it arrive?

430. Mia drives 180 km at 90 km/h, stops for 30 minutes, then drives another 60 km at 80 km/h.

(a) How long does the first part of the trip take? 1 mark

(b) How long does the last part of the trip take, in minutes? 1 mark

(c) Find the total time for the trip and Mia's average speed for the whole trip, correct to one decimal place. 2 marks

Revision

431. Which is the better buy: 2 kg of apples for \$7.40, or 3 kg for \$10.80?

432. A 600 g pack costs \$4.20. Find the price per 100 g.

433. Which is the better buy: 12 cans for \$9.00, or 30 cans for \$21.90?

434. Convert 81 km/h to m/s.

435. Convert 7 m/s to km/h.

436. Convert 400 mL/min to L/h.

437. 8 kg of potatoes cost \$72. Find the cost per kg.

438. Is "5 km to 2 km" a ratio or a rate?

439. Simplify 0.6 : 0.9.

440. Share 360 g of flour in the ratio 1 : 2 : 3.

441. A map has a scale of 1 : 50 000. Two towns are 9 cm apart on the map. How far apart are they in km?

442. A plan has a scale of $1 : 50$. A bench is 8 cm long on the plan. How long is the real bench in metres?

443. Find 7% of \$900.

444. Decrease \$3200 by 15%.

445. A price including 10% GST is \$55. Find the price before GST.

446. Find the pay for 4 hours at time-and-a-half when the normal rate is \$30 per hour.

447. Write 1.75 hours in hours and minutes.

448. Convert 3500 m to kilometres.

449. Convert 0.2 L to millilitres.

450. Solve $2x + 9 = 31$.

451. Use $V = lwh$ to find V when $l = 4$, $w = 2.5$ and $h = 2$.

452. Find the circumference of a circle with radius 10 cm, correct to one decimal place.

453. Find the mean of $50,\ 60,\ 70,\ 80,\ 90$.

454. Find the range of $2.5,\ 7.1,\ 4.8$.

455. A letter is chosen at random from A, B, C, D, E. What is the probability it is a vowel?

456. Round 0.985 to one decimal place.

1.10 Fuel consumption

SYLLABUS MST-12-S1-05

- Examine fuel consumption rates to compare the efficiency of vehicles

NOTES

- **Fuel consumption** is the number of litres a vehicle uses to travel 100 km, written in L/100 km.
- Fuel consumption $= \dfrac{\text{litres used}}{\text{km travelled}} \times 100$: 36 L for 400 km is $\frac{36}{400} \times 100 = 9$ L/100 km.
- Fuel needed = fuel consumption × number of lots of 100 km: 8 L/100 km for 350 km is $8 \times 3.5 = 28$ L.
- Distance on a tank $= \dfrac{\text{litres in tank}}{\text{fuel consumption}} \times 100$.
- A **lower** fuel consumption means a **more efficient** vehicle. Fuel cost = litres × price per litre.

WORKED EXAMPLE

A car uses 40 L of fuel to travel 500 km. Find its fuel consumption in L/100 km.

$\frac{40}{500} \times 100$ — litres ÷ km × 100

$= 8$ L/100 km — litres per 100 km

Tier 1 · Find the fuel consumption

457. A car uses 27 L of fuel to travel 300 km. Find its fuel consumption.

458. A ute uses 60 L of fuel to travel 750 km. Find its fuel consumption.

459. A van uses 44 L of fuel to travel 400 km. Find its fuel consumption.

Tier 2 · Decimal values

460. A car uses 33 L to travel 600 km. Find its fuel consumption.

461. A car uses 48.6 L to travel 540 km. Find its fuel consumption.

462. A car uses 25.2 L to travel 360 km. Find its fuel consumption.

Tier 3 · Fuel needed for a trip

463. A car uses 6.5 L/100 km. How much fuel does it need for an 800 km trip?

464. A 4WD uses 11 L/100 km. How much fuel does it need for a 250 km trip?

465. A car uses 7.2 L/100 km. How much fuel does it need to travel 1500 km?

Tier 4 · Fuel cost

466. A car uses 8 L/100 km. Petrol costs $1.90 per litre. Find the fuel cost for a 600 km trip.

467. A small car uses 5.5 L/100 km. Petrol costs $2.05 per litre. Find the fuel cost for a 320 km trip.

468. A 4WD uses 12 L/100 km. Diesel costs $1.98 per litre. Find the fuel cost for a 450 km trip.

Tier 5 · Distance on a tank

469. A car uses 7.5 L/100 km and has a 60 L tank. How far can it travel on a full tank?

470. A car uses 9 L/100 km and has a 45 L tank. How far can it travel on a full tank?

471. A ute uses 12.5 L/100 km and has a 70 L tank. How far can it travel on a full tank?

Tier 6 · Comparing efficiency

472. Car A uses 32 L to travel 400 km. Car B uses 39 L to travel 600 km. Which car is more efficient?

473. A ute uses 66 L to travel 550 km. A van uses 70 L to travel 625 km. Which is more efficient?

474. A hybrid uses 18 L to travel 450 km. A small car uses 30 L to travel 500 km. Which is more efficient?

Tier 7 · Savings over time

475. A driver travels 20 000 km a year. Petrol costs \$2.10 per litre. How much would they save each year by driving a car that uses 5 L/100 km instead of one that uses 8 L/100 km?

476. A driver travels 12 000 km a year. Petrol costs \$1.95 per litre. How much would they save each year with a car using 6.5 L/100 km instead of 9.5 L/100 km?

477. Jade drives 300 km a week. Her car uses 10 L/100 km. A new car would use 7 L/100 km. Petrol costs \$2.00 per litre. How much would she save in a year (52 weeks)?

Conceptual

478. Explain why a lower fuel consumption means a more efficient vehicle.

479. Jo says a car that uses 40 L to travel 400 km has a fuel consumption of 0.1 L/100 km. Explain her mistake.

480. Explain why vehicles are compared using L/100 km instead of the total fuel used.

481. A car's fuel consumption is higher in city driving than on the highway. Give one reason why.

482. A car uses 7.8 L/100 km and has a 52 L tank.

(a) How much fuel does it need for a 450 km trip? 1 mark

(b) Petrol costs \$1.92 per litre. Find the cost of the fuel for this trip. 1 mark

(c) Can the car travel 700 km on one full tank? Justify your answer. 2 marks

Revision

483. How long does it take to travel 120 km at 80 km/h?

484. A car travels at 65 km/h for 2.4 hours. How far does it travel?

485. A bus travels 270 km in 3 hours 45 minutes. Find its average speed.

486. How long does it take to travel 140 km at 84 km/h? Give your answer in hours and minutes.

487. Which is the better buy: 1 L of milk for \$3.20, or 3 L for \$9.30?

488. An 800 g pack costs \$5.60. Find the price per 100 g.

489. Convert 36 km/h to m/s.

490. Convert 4 m/s to km/h.

491. Convert 15 L/h to mL/min.

492. Zac earns \$133 for 7 hours of work. Find his hourly rate.

493. Simplify $40 : 64$.

494. Share \$45 in the ratio $2 : 7$.

495. A map has a scale of $1 : 200\,000$. Two towns are 6 cm apart on the map. How far apart are they?

496. On a $1 : 100$ plan, a room is 5 cm by 3.6 cm. Find its real area.

497. Find 30% of \$85.

498. Increase \$64 by 12.5%.

499. Leah works 40 hours a week at \$26.50 per hour. Find her weekly pay.

500. Write 0.75 hours in minutes.

501. Write 2 hours 50 minutes in hours, correct to two decimal places.

502. Convert 5.4 km to metres.

503. Solve $\frac{x}{3} + 2 = 8$.

504. A square has an area of 49 m^2. Find its perimeter.

505. Find the mean of $1.2,\ 1.8,\ 2.4$.

506. Find the median of $4,\ 9,\ 2,\ 7,\ 6,\ 1$.

507. A standard die is rolled. What is the probability of rolling a number greater than 4?

508. Round 15.449 to one decimal place.

Solutions

1. $5 : 23$

2. $6 : 1$

3. $3 : 2$

4. $9 : 7$

5. $11 : 4$

6. $5 : 2$

7. $2 : 3 : 1$

8. $1 : 2 : 3$

9. $5 : 4 : 7$

10. $3 + 4 = 7$ pens, so $\frac{3}{7}$ are red.

11. $5 + 2 = 7$ days, so $\frac{2}{7}$ were rainy.

12. $9 + 4 = 13$ people, so $\frac{4}{13}$ are children.

13. $2 + 5 = 7$ parts, so $\frac{5}{7}$ are dogs.

14. $4 + 1 = 5$ parts, so $\frac{1}{5}$ is cement.

15. $7 + 2 = 9$ parts, so the team won $\frac{7}{9}$ of its games.

16. $\frac{3}{8}$

17. $\frac{1}{5}$

18. $\frac{2}{9}$

19. Non-bus $= 8 - 3 = 5$ parts, so the ratio is $3 : 5$.

20. Garden beds $= 4 - 1 = 3$ parts, so the ratio is $1 : 3$.

21. Backs $= 9 - 5 = 4$ parts, so the ratio is $5 : 4$.

22. There are $3 + 5 = 8$ parts in total, so apples are $\frac{3}{8}$ of the fruit. The fraction $\frac{3}{5}$ compares apples to oranges, not apples to all the fruit.

23. No. Order matters. $2 : 7$ means 2 boys for every 7 girls (more girls). $7 : 2$ would mean 7 boys for every 2 girls (more boys).

24. Boys $= 12 - 5 = 7$ parts, so girls : boys $= 5 : 7$.

25. (a) $5 + 3 = 8$ parts, so the team lost $\frac{3}{8}$. (b) Yes. The team won $\frac{5}{8}$, and $\frac{5}{8} > \frac{4}{8} = \frac{1}{2}$.

26. (a) $5 : 2$. (b) $2 + 5 = 7$ parts, so $\frac{2}{7}$ are tea. (c) The total is 7 parts, not 5. The fraction $\frac{2}{5}$ compares tea to coffee, not tea to all the drinks sold.

27. \$34

28. $0.25 \times 86 = \$21.50$

29. $60 \times 1.15 = \$69$

30. $250 \times 0.8 = \$200$

31. $24.50 \times 18 = \$441$

32. $22 \times 1.5 \times 4 = \$132$

33. $3.5 \times 100 = 350$ cm

34. $2400 \div 1000 = 2.4$ kg

35. $750 \div 1000 = 0.75$ L

36. $1.2 \times 1000 = 1200$ m

37. $C = 5 \times 8 + 20 = 60$, so \$60.

38. $A = 12 \times 7.5 = 90$

39. $x = 14$

40. $x = 9$

41. $2 \times (8 + 5) = 26$ m

42. $8 \times 5 = 40$ m^2

43. 9:45 am to 1:45 pm is 4 h, then 35 min more: 4 h 35 min.

44. $0.25 \times 60 = 15$, so 2 h 15 min.

45. $35 \div 5 = 7$

46. In order: 2, 3, 6, 8, 9. Median = 6.

47. $19 - 5 = 14$

48. $\frac{1}{6}$

49. $\frac{3}{3+7} = \frac{3}{10}$

50. $45 \times 1.1 = \$49.50$

51. $35 \times 12 = \$420$

52. 3.48

53. 2 : 5

54. 3 : 2

55. 3 : 4

56. HCF = 12, so 2 : 3.

57. HCF = 9, so 2 : 5.

58. HCF = 6, so 5 : 8.

59. Divide by 3: 2 : 3 : 5.

60. Divide by 4: 3 : 4 : 5.

61. Divide by 7: 2 : 1 : 3.

62. $12 : 20 = 3 : 5$

63. $40 : 16 = 5 : 2$

64. $250 : 400 = 5 : 8$

65. 2 h = 120 min, so $30 : 120 = 1 : 4$.

66. 1.2 kg = 1200 g, so $400 : 1200 = 1 : 3$.

67. 2 L = 2000 mL, so $600 : 2000 = 3 : 10$.

68. $\times 10$: $6 : 15 = 2 : 5$.

69. $\times 10$: $25 : 40 = 5 : 8$.

70. $\times 10$: $12 : 8 = 3 : 2$.

71. $\times 6$: $2 : 3$.

72. $\times 10$: $4 : 3$.

73. $\times 4$: $3 : 2$.

74. No. 2 still divides into both 6 and 8. The simplest form is $3 : 4$ (divide $12 : 16$ by the HCF, 4).

75. Yes. $4 : 6 = 2 : 3$ and $10 : 15 = 2 : 3$, so they have the same simplest form.

76. $8 = 2 \times 4$, so the second number is $5 \times 4 = 20$.

77. 1.5 L = 1500 mL, so $500 : 1500 = 1 : 3$.

78. (a) $18 : 12 = 3 : 2$. (b) $12 + 2 = 14$ boys, so $18 : 14 = 9 : 7$. (c) 3 still divides into both 9 and 6, so it simplifies further to $3 : 2$.

79. $5 : 8$

80. $\frac{3}{7}$

81. $2 : 7$

82. $4 + 1 + 3 = 8$

83. \$12

84. \$75

85. $120 \times 1.1 = \$132$

86. $90 \times 0.75 = \$67.50$

87. $\frac{18}{24} \times 100 = 75\%$

88. $26 \times 35 = \$910$

89. $30 \times 2 \times 3 = \$180$

90. $62\,400 \div 52 = \$1200$

91. 4200 g

92. 3.5 m

93. 2500 mL

94. $90 \div 60 = 1.5$ h

95. $P = 18 + 8 = 26$

96. $C = 40 + 45 = 85$, so \$85.

97. $x = 19$

98. $x = 30$

99. $12 \times 4.5 = 54$ cm^2

100. $50 \div 4 = 12.5$

101. 5

102. In order: 2, 4, 7, 9. Median $= \dfrac{4+7}{2} = 5.5$.

103. $\dfrac{5}{8}$

104. 10:50 am to 12:50 pm is 2 h, then 25 min more: 2 h 25 min.

105. 5 parts, one part = \$6. Zoe \$12, Liam \$18.

106. 6 parts, one part = \$8. Ava \$8, Noah \$40.

107. 7 parts, one part = 8. 24 and 32 lollies.

108. 11 parts, one part = \$12. \$84 and \$48.

109. 8 parts, one part = \$45. \$225 and \$135.

110. 5 parts, one part = 125 g. 375 g and 250 g.

111. 6 parts, one part = \$20. \$20, \$40 and \$60.

112. 9 parts, one part = 20 L. 40 L, 60 L and 80 L.

113. 10 parts, one part = \$50. Ali \$150, Bec \$250, Cam \$100.

114. One part $= 14 \div 2 = 7$. Dogs $= 5 \times 7 = 35$.

115. One part $= 3$ bags. Sand $= 4 \times 3 = 12$ bags.

116. One part $= 24 \div 3 = 8$. Adults $= 5 \times 8 = 40$.

117. 5 parts, one part $= 0.5$ kg. Cement 0.5 kg, sand 2 kg.

118. 26 parts, one part $= 0.1$ L. Oil $= 0.1$ L $= 100$ mL.

119. 5 parts, one part $= 0.8$ L. Red 1.6 L, white 2.4 L.

120. One part $= \$150$. Jai \$150, Ruby \$450. Ruby gets \$300 more.

121. One part $= 8$. Girls 40, boys 32. There are 8 more girls.

122. One part $= \$300$. Shares \$900 and \$1500. Difference $= \$600$.

123. Difference $= 5 - 2 = 3$ parts $= \$60$, so one part $= \$20$. Total $= 7 \times \$20 = \$140$.

124. 3 parts $= 15$ kg, so one part $= 5$ kg. Total $= 11 \times 5 = 55$ kg.

125. 3 parts $= \$36$, so one part $= \$12$. Mel \$12, Dan \$48.

126. $\$3 + \$5 = \$8$, not \$40. There are 8 parts, so one part $= \$5$. The shares are \$15 and \$25.

127. No. Pia is first in the ratio, so she gets 2 parts: $2 \times \$10 = \$20$. Rob gets \$30. Sam swapped the shares.

128. No. There are 7 parts and $\$50 \div 7$ is not a whole number (about \$7.14), so the shares would be about \$21.43 and \$28.57.

129. Ratio 10 : 12 : 14, so 36 parts and one part $= \$250$. Eldest $= 14 \times \$250 = \$3500$.

130. (a) 6 parts, so $\frac{1}{6}$ is cement. (b) One part $= 48 \div 6 = 8$ kg, so sand $= 5 \times 8 = 40$ kg. (c) One part $= 15$ kg, so sand $= 5 \times 15 = 75$ kg.

131. 7 : 2

132. $\frac{9}{11}$

133. 3 : 7

134. 3 : 4

135. 5 : 3

136. 2 : 3 : 5

137. $45 : 120 = 3 : 8$

138. $4 : 18 = 2 : 9$

139. $\times 12$: $3 : 8$

140. \$13

141. $80 \times 1.05 = \$84$

142. $150 \times 0.6 = \$90$

143. $28.40 \times 7.5 = \$213$

144. $24 \times 1.5 \times 6 = \$216$

145. $250 \times 1.1 = \$275$

146. 3750 m

147. 850 mL

148. 6.5 kg

149. $v = 4 + 15 = 19$

150. $3x = 15$, so $x = 5$.

151. $\frac{1}{2} \times 10 \times 6 = 30 \text{ cm}^2$

152. $4 \times 7.5 = 30$ m

153. $50 \div 5 = 10$

154. $31 - 8 = 23$

155. $\dfrac{13}{52} = \dfrac{1}{4}$

156. 12.7

157. $9 \times 50 = 450$ cm

158. $2.5 \times 300 = 750$ cm

159. $6 \times 400 = 2400$ cm

160. $8 \times 1000 = 8000 \text{ cm} = 80$ m

161. $4.2 \times 500 = 2100$ cm $= 21$ m

162. $3 \times 2000 = 6000$ cm $= 60$ m

163. $6 \times 100\,000 = 600\,000$ cm $= 6$ km

164. $8.4 \times 25\,000 = 210\,000$ cm $= 2.1$ km

165. $3.2 \times 250\,000 = 800\,000$ cm $= 8$ km

166. 9 m $= 900$ cm, $\quad 900 \div 100 = 9$ cm

167. 4 km $= 400\,000$ cm, $\quad 400\,000 \div 50\,000 = 8$ cm

168. 1.5 km $= 150\,000$ cm, $\quad 150\,000 \div 20\,000 = 7.5$ cm

169. 10 m $= 1000$ cm, so $2 : 1000 = 1 : 500$.

170. 1 km $= 100\,000$ cm, so $5 : 100\,000 = 1 : 20\,000$.

171. 15 km $= 1\,500\,000$ cm, so $3 : 1\,500\,000 = 1 : 500\,000$.

172. $2.5 \times 200 = 500$ m

173. $2.5 + 4 = 6.5$ cm, $\quad 6.5 \times 200 = 1300$ m $= 1.3$ km

174. $2.5 + 4 + 3.5 = 10$ cm, $\quad 10 \times 200 = 2000$ m $= 2$ km

175. Both parts of the scale are in the same unit. 1 cm represents 1000 cm, which is 10 m, not 1000 m.

176. $1 : 10\,000$. On it, 1 cm represents 100 m. On the other, 1 cm represents 10 km, so a lot more land is squeezed into each centimetre.

177. 12 km $= 1\,200\,000$ cm, $\quad 1\,200\,000 \div 200\,000 = 6$ cm

178. $18 \times 500 = 9000$ cm $= 90$ m. No, the oval is only 90 m long.

179. (a) $6 \times 25\,000 = 150\,000$ cm $= 1.5$ km. (b) 4 km $= 400\,000$ cm, $\quad 400\,000 \div 25\,000 = 16$ cm. (c) $8 : 400\,000$, divide by 8: $1 : 50\,000$.

180. $4 : 5$

181. $5 : 200 = 1 : 40$

182. $15 : 45 = 1 : 3$

183. One part = \$10: \$10 and \$50.

184. One part $= \$40$: \$160 and \$200.

185. One part $= 12$ kg: 12 kg, 24 kg and 36 kg.

186. One part $= 6$, so boys $= 12$.

187. $\dfrac{9}{13}$

188. $2:5$

189. \$84

190. $45 \times 1.2 = \$54$

191. $96 \times 0.875 = \$84$

192. $31.20 \times 12 = \$374.40$

193. $25 \times 2 \times 4 = \$200$

194. 650 m

195. 480 cm

196. 2.5 L

197. $2x = 18$, so $x = 9$.

198. $C = 25 + 60 = 85$, so \$85.

199. $6.5 \times 4 = 26\ \text{m}^2$

200. $80 \div 4 = 20$

201. In order: 3, 5, 7, 8, 9, 12. Median $= \dfrac{7+8}{2} = 7.5$.

202. $\dfrac{3}{6} = \dfrac{1}{2}$

203. $3 \times 100 = 300$ cm $= 3$ m and $4 \times 100 = 400$ cm $= 4$ m. The kitchen is 3 m by 4 m.

204. 6 m by 3.5 m.

205. $3 \times 50 = 150$ cm $= 1.5$ m and $1.6 \times 50 = 80$ cm $= 0.8$ m.

206. $3 \times 4 = 12\ \text{m}^2$

207. $6 \times 3.5 = 21\ \text{m}^2$

208. $4.5\ \text{m} \times 3\ \text{m} = 13.5\ \text{m}^2$

209. $12 \times \$28 = \$336$

210. $21 \times \$18 = \$378$

211. $13.5 \times \$55 = \$742.50$

212. 4 (two in the living room, one in the kitchen, one in the bedroom).

213. The living room.

214. 5: living room, kitchen, bedroom, bathroom and the front door.

215. $12 + 20 = 32\ \text{m}^2$, $32 \times \$40 = \$1280$

216. $12 + 3 = 15\ \text{m}^2$, $15 \times \$35 = \$525$

217. The house is 8 cm by 7 cm on the plan, so 8 m by 7 m: $8 \times 7 = 56\ \text{m}^2$.

218. One tile $= 0.3 \times 0.3 = 0.09\ \text{m}^2$. $3 \div 0.09 = 33.3\ldots$, so 34 tiles.

219. Wall area $= 14 \times 2.4 = 33.6\ \text{m}^2$. $33.6 \div 16 = 2.1$, so 3 L.

220. $6 \times 200 = 1200$ cm $= 12$ m and $4 \times 200 = 800$ cm $= 8$ m. Area $= 96\ \text{m}^2$. $96 \div 1.5 = 64$ rolls.

221. Carpet $= \$624$. Vinyl $= \$468 + \$120 = \$588$. Vinyl is cheaper by \$36.

222. Shop A $= 12 \times \$30 = \$360$. Shop B: $12 \div 1.5 = 8$ boxes, $8 \times \$42 = \$336$. Shop B is cheaper by \$24.

223. Two 4 L tins $= \$136$. Six 1 L tins $= \$132$. Six 1 L tins is cheaper by \$4.

224. Both the length and the width are 100 times bigger, so the area is not just 100 times bigger. Change the lengths first: $4\ \text{m} \times 3\ \text{m} = 12\ \text{m}^2$.

225. 33 tiles would not cover the whole floor. You can't buy part of a tile, so round up to 34.

226. $8 \times 50 = 400$ cm $= 4$ m and $6 \times 50 = 300$ cm $= 3$ m. Area $= 12\ \text{m}^2$.

227. $12 \times 1.1 = 13.2$, so round up to 14 tiles.

228. (a) 6 m by 4 m. (b) $6 \times 4 = 24\ \text{m}^2$. (c) Boards $= 24 \times \$65 = \$1560$. Stain: $24 \div 10 = 2.4$, so 3 tins $= \$135$. Total $= \$1695$.

229. $3 : 4$

230. $80 : 200 = 2 : 5$

231. One part $= \$12$: \$60 and \$36.

232. One part $= 0.5$ L: 0.5 L and 1 L.

233. $4 \times 25\,000 = 100\,000$ cm $= 1$ km

234. $3\,000\,000 \div 200\,000 = 15$ cm

235. $2 : 500\,000 = 1 : 250\,000$

236. $3 : 2$

237. $\dfrac{4}{11}$

238. \$54

239. $250 \times 1.08 = \$270$

240. $132 \div 1.1 = \$120$

241. $38 \times 29.50 = \$1121$

242. $32 \times 1.5 \times 3 = \$144$

243. 1.25 m

244. 800 mm

245. 2750 mL

246. $2 \times (7.2 + 3.5) = 21.4$ m

247. $\frac{1}{2} \times 8 \times 5 = 20$ m^2

248. $\pi \times 3^2 \approx 28.3$ m^2

249. $5x = 35$, so $x = 7$.

250. $d = 150$

251. $40 \div 5 = 8$

252. 6

253. $\frac{4}{12} = \frac{1}{3}$

254. 0.05

255. $240 \div 4 = 60$ km/h

256. $1500 \div 5 = 300$ m/min

257. $45 \div 9 = 5$ L/min

258. $217 \div 7 = \$31$/h

259. $84 \div 12 = \$7$/m

260. $1350 \div 5 = \$270$/day

261. $13.80 \div 4 = \$3.45$/kg

262. $10.50 \div 6 = \$1.75$/L

263. $27 \div 8 = 3.375$, so \$3.38/m

264. 1 min $= 4 \times 15$ s, so $18 \times 4 = 72$ beats/min.

265. 1 min $= 2 \times 30$ s, so $250 \times 2 = 500$ mL/min.

266. 1 h $= 4 \times 15$ min, so $20 \times 4 = 80$ km/h.

267. $21.25 \div 2.5 = \$8.50$/kg

268. $4.20 \div 1.5 = \$2.80$/L

269. $6 \div 0.4 = \$15$/kg

270. 3.5 h: $315 \div 3.5 = 90$ km/h

271. 2.5 h: $57 \div 2.5 = \$22.80$/h

272. 45 min $= 0.75$ h, so 2.75 h: $220 \div 2.75 = 80$ km/h

273. A ratio compares quantities in the same units, e.g. 2 cups of cordial to 7 cups of water, 2 : 7. A rate compares quantities in different units, e.g. \$25 per hour.

274. A ratio, because both quantities are measured in the same unit (cups).

275. He divided the wrong way. km/h means km divided by hours: $200 \div 4 = 50$ km/h.

276. $195 \div 3 = \$65/\text{day}$. $5 \text{ days} = 5 \times \$65 = \$325$.

277. (a) $36 \div 4 = 9$ L/h. (b) $9 \times 24 = 216$ L. (c) $216 \times 7 = 1512$ L. $1512 \div 1000 \times \$3.20 = \$4.84$.

278. 3 : 4

279. 250 : 1000 = 1 : 4

280. One part = \$20: \$60 and \$80.

281. One part = 5, so girls = 25.

282. $7 \times 50\,000 = 350\,000$ cm $= 3.5$ km

283. $300\,000 \div 20\,000 = 15$ cm

284. $4.5 \text{ m} \times 3 \text{ m} = 13.5 \text{ m}^2$

285. 6

286. $15 \times \$38 = \$570$

287. $\frac{2}{9}$

288. \$60

289. $75 \times 1.4 = \$105$

290. $71\,500 \div 26 = \$2750$

291. $27 \times 2 \times 5 = \$270$

292. 3600 m

293. 4.25 m

294. 1 h 35 min

295. $x = 36$

296. $A = 42$

297. $2.5 \times 2.5 = 6.25 \text{ m}^2$

298. $52 \div 4 = 13$

299. $41 - 17 = 24$

300. $\frac{4}{52} = \frac{1}{13}$

301. 4000 m/min

302. 2500 mL/h

303. 300 c/min

304. $180 \div 60 = 3$ km/min

305. $6 \times 60 = 360$ m/min

306. $1200 \div 60 = 20$ L/min

307. $\frac{72\,000}{3600} = 20$ m/s

308. $\frac{36\,000}{3600} = 10$ m/s

309. $\frac{108\,000}{3600} = 30$ m/s

310. $5 \times 3600 = 18\,000$ m $= 18$ km, so 18 km/h.

311. $25 \times 3600 = 90\,000$ m $= 90$ km, so 90 km/h.

312. $12 \times 3600 = 43\,200$ m $= 43.2$ km, so 43.2 km/h.

313. $250 \times 60 = 15\,000$ mL $= 15$ L, so 15 L/h.

314. $800 \times 60 = 48\,000$ mL $= 48$ L, so 48 L/h.

315. $75 \times 60 = 4500$ mL $= 4.5$ L, so 4.5 L/h.

316. $6000 \div 60 = 100$ mL/min

317. $1800 \div 60 = 30$ mL/min

318. $12\,000 \div 60 = 200$ mL/min

319. $10.4 \times 3.6 = 37.44$ km/h, which is less than 40 km/h. The e-scooter is faster.

320. Tap B: $540 \div 60 = 9$ L/min. Tap A is faster.

321. Drip B: $150 \div 60 = 2.5$ mL/min. Drip B is faster.

322. Changing km to m multiplies by 1000 and changing hours to seconds divides by 3600. $1000 \div 3600 = \frac{1}{3.6}$, so the two steps together divide by 3.6.

323. Yes. In one hour the object travels $20 \times 3600 = 72\,000$ m, which is 72 km.

324. $15 \times 3.6 = 54$ km/h, which is more than 50 km/h. Yes, the car is speeding.

325. $125 \div 60 \approx 2.1$ mL/min

326. (a) $18 \times 60 = 1080$ L/h. (b) $18\,000 \div 60 = 300$ mL/s. (c) $2700 \div 18 = 150$ min $= 2$ h 30 min.

327. A rate, because it compares quantities in different units (km and hours).

328. 70 km/h

329. \$13/h

330. \$7.50/kg

331. $24 \times 3 = 72$ beats/min

332. $260 \div 3.25 = 80$ km/h

333. 2 : 3

334. \$100 and \$150

335. $80\,000$ cm $= 800$ m

336. $5 \times 3.2 = 16$ m^2

337. 4

338. $\frac{5}{8}$

339. \$77

340. \$48

341. \$75 400

342. $26 \times 1.5 \times 6 =$ \$234

343. 350 mL

344. 7200 m

345. 2.5 min

346. $4x = 28$, so $x = 7$.

347. $P = 21$

348. $3 \times 2 \times 1.5 = 9\ \text{m}^3$

349. $35 \div 5 = 7$

350. In order: 8, 10, 13, 15, 21. Median $= 13$.

351. $\frac{5}{6}$

352. 1235

353. \$9.90/kg vs \$9.50/kg. The 3 kg pack.

354. \$1.60/L vs \$1.70/L. The 4 L bottle.

355. \$0.75 vs \$0.70 per roll. The 9-pack.

356. $4.20 \div 0.6 = \$7.00$/kg vs $9.75 \div 1.5 = \$6.50$/kg. The 1.5 kg bag.

357. \$4.80/L vs \$4.40/L. The 1.25 L bottle.

358. \$6.00/L vs \$5.20/L. The 1 L carton.

359. $2.70 \div 1.5 = \$1.80$ vs $6.40 \div 4 = \$1.60$ per 100 g. The 400 g bag.

360. \$1.80 vs \$1.90 per 100 g. The 200 g pack.

361. \$1.70 vs \$1.60 per 100 g. The 1 kg pack.

362. \$0.80, \$0.75, \$0.80 each. The 10-pack.

363. \$2.80, \$2.60, \$2.70 per L. The 2 L bottle.

364. \$2.50, \$2.30, \$2.35 per 100 g. The 250 g box.

365. Amir \$24/h, Bella \$23/h. Amir.

366. A: 30 L/min. B: $1500 \div 60 = 25$ L/min. Pump A.

367. A: $5 \div 25 = 0.2$ km/min. B: $8 \div 44 \approx 0.18$ km/min. Runner A.

368. 1 kg: $15 \times 0.8 = \$12$/kg. 1.5 kg: $17.40 \div 1.5 = \$11.60$/kg. The 1.5 kg pack.

369. 2 L: $\$3 \div 2 = \$1.50$/L. 3 L: \$1.40/L. The 3 L bottle.

370. 500 g: $7 \times 0.9 = \$6.30$, so \$1.26 per 100 g. 750 g: \$1.20 per 100 g. The 750 g tub.

371. \$4.90/kg vs \$5.10/kg. No, the 2 kg bag is the better buy.

372. Per 100 mL: $1.80 \div 3.75 = \$0.48$ vs $6.25 \div 12.5 = \$0.50$. The can.

373. \$0.55 vs \$0.58 per egg. The dozen.

374. The packs are different sizes, so the prices are for different amounts. The unit price gives the cost of the same amount of each, so they can be compared fairly.

375. For example, 2 kg of sugar for \$9.80 is \$4.90/kg, but 5 kg for \$25.50 is \$5.10/kg. The smaller bag is cheaper per kg.

376. $1.20 \times 4.5 = \$5.40$

377. For example: they only need a small amount and the rest would be wasted or go off; they cannot afford the bigger pack right now; they have no room to store it.

378. (a) $5.25 \div 7.5 = \$0.70$. (b) $8.16 \div 12 = \$0.68$. (c) Four packs of X give exactly 3 kg for $4 \times \$5.25 = \$21.00$. Three packs of Y give 3.6 kg for \$24.48, and two Y plus one X give 3.15 kg for \$21.57. Four packs of Brand X is cheapest, even though Y has the lower unit price.

379. $54 \div 3.6 = 15$ m/s

380. $8 \times 3.6 = 28.8$ km/h

381. $300 \times 60 = 18\,000$ mL $= 18$ L/h

382. $9000 \div 60 = 150$ mL/min

383. 70 km/h

384. \$27/h

385. A ratio (same units).

386. $12 : 36 = 1 : 3$

387. 36 and 48

388. $250\,000$ cm $= 2.5$ km

389. $3.5 \times 4 = 14$ m^2

390. 3

391. \$81

392. $1200 \times 1.025 = \$1230$

393. \$38

394. $1140 \div 38 = \$30/\text{h}$

395. $28 \times 2 \times 7.5 = \$420$

396. 150 min

397. 450 g

398. $x + 2 = 7$, so $x = 5$.

399. $2 \times \pi \times 5 \approx 31.4$ cm

400. $36 \div 9 = 4$ m

401. $108 \div 4 = 27$

402. 3

403. $\frac{1}{2}$

404. 8.00

405. $70 \times 3 = 210$ km

406. $18 \times 2.5 = 45$ km

407. $5 \times 1.5 = 7.5$ km

408. $300 \div 75 = 4$ h

409. $36 \div 12 = 3$ h

410. $9 \div 6 = 1.5$ h $= 1$ h 30 min

411. $225 \div 2.5 = 90$ km/h

412. 45 min $= 0.75$ h: $12 \div 0.75 = 16$ km/h

413. $400 \div 50 = 8$ m/s

414. $100 \div 80 = 1.25$ h $= 1$ h 15 min

415. $50 \div 60 = 0.833\ldots$ h, $\quad 0.833\ldots \times 60 = 50$ min

416. $245 \div 70 = 3.5$ h $= 3$ h 30 min

417. 90 km

418. 3 h

419. $240 \div 4 = 60$ km/h

420. 50 km in 1 h: 50 km/h

421. From 1 h to 1.5 h: 30 minutes

422. $140 - 50 = 90$ km in $3 - 1.5 = 1.5$ h: $90 \div 1.5 = 60$ km/h

423. A: $100 \div 2 = 50$ km/h. B: $100 \div 2.5 = 40$ km/h. A is faster.

424. A has ridden 100 km and B has ridden 80 km, so they are 20 km apart.

425. $2.5 - 2 = 0.5$ h $= 30$ minutes

426. The object has stopped. Time is passing but the distance is not changing.

427. The steeper line travels further in the same time, so it shows the faster object.

428. 0.5 h is half an hour, which is $0.5 \times 60 = 30$ min, not 50 min. The trip takes 2 h 30 min.

429. $210 \div 84 = 2.5$ h $= 2$ h 30 min. It arrives at 11:45 am.

430. (a) $180 \div 90 = 2$ h. (b) $60 \div 80 = 0.75$ h $= 45$ min. (c) Total time $= 2 + 0.5 + 0.75 = 3.25$ h. Average speed $= 240 \div 3.25 \approx 73.8$ km/h.

431. \$3.70/kg vs \$3.60/kg. The 3 kg bag.

432. $4.20 \div 6 =$ \$0.70

433. \$0.75 vs \$0.73 per can. The 30-pack.

434. $81 \div 3.6 = 22.5$ m/s

435. $7 \times 3.6 = 25.2$ km/h

436. 24 L/h

437. \$9/kg

438. A ratio (same units).

439. $6 : 9 = 2 : 3$

440. 60 g, 120 g and 180 g

441. 4.5 km

442. $8 \times 50 = 400$ cm $= 4$ m

443. \$63

444. $3200 \times 0.85 = \$2720$

445. $55 \div 1.1 = \$50$

446. $30 \times 1.5 \times 4 = \$180$

447. 1 h 45 min

448. 3.5 km

449. 200 mL

450. $2x = 22$, so $x = 11$.

451. $V = 20$

452. $2 \times \pi \times 10 \approx 62.8$ cm

453. 70

454. $7.1 - 2.5 = 4.6$

455. $\frac{2}{5}$

456. 1.0

457. $\frac{27}{300} \times 100 = 9$ L/100 km

458. $\frac{60}{750} \times 100 = 8$ L/100 km

459. $\frac{44}{400} \times 100 = 11$ L/100 km

460. $\frac{33}{600} \times 100 = 5.5$ L/100 km

461. $\frac{48.6}{540} \times 100 = 9$ L/100 km

462. $\frac{25.2}{360} \times 100 = 7$ L/100 km

463. $6.5 \times 8 = 52$ L

464. $11 \times 2.5 = 27.5$ L

465. $7.2 \times 15 = 108$ L

466. $8 \times 6 = 48$ L, $48 \times \$1.90 = \$91.20$

467. $5.5 \times 3.2 = 17.6$ L, $17.6 \times \$2.05 = \$36.08$

468. $12 \times 4.5 = 54$ L, $54 \times \$1.98 = \$106.92$

469. $60 \div 7.5 = 8$ lots of 100 km $= 800$ km

470. $45 \div 9 = 5$ lots of 100 km $= 500$ km

471. $70 \div 12.5 = 5.6$ lots of 100 km $= 560$ km

472. A: 8 L/100 km. B: 6.5 L/100 km. Car B.

473. Ute: 12 L/100 km. Van: 11.2 L/100 km. The van.

474. Hybrid: 4 L/100 km. Small car: 6 L/100 km. The hybrid.

475. 200 lots of 100 km. $5 \times 200 = 1000$ L and $8 \times 200 = 1600$ L. Saves 600 L $\times \$2.10 = \$1260$.

476. 120 lots of 100 km. Difference $= 3 \times 120 = 360$ L. Saves $360 \times \$1.95 = \$702$.

477. Difference $= 3 \times 3 = 9$ L a week $= \$18$ a week. $\$18 \times 52 = \$936$ a year.

478. It means the vehicle uses fewer litres to travel the same 100 km, so it costs less to run.

479. $40 \div 400 = 0.1$ is the litres per 1 km. She forgot to multiply by 100: $0.1 \times 100 = 10$ L/100 km.

480. The vehicles may have travelled different distances. L/100 km compares them over the same distance.

481. For example: stopping and starting at traffic lights, idling in traffic, or driving at low speeds uses more fuel per km.

482. (a) $7.8 \times 4.5 = 35.1$ L. (b) $35.1 \times \$1.92 = \$67.39$. (c) $52 \div 7.8 \times 100 \approx 667$ km. No, a full tank only gets it about 667 km.

483. 1.5 h = 1 h 30 min

484. 156 km

485. $270 \div 3.75 = 72$ km/h

486. $140 \div 84 = 1.666\ldots$ h = 1 h 40 min

487. \$3.20/L vs \$3.10/L. The 3 L bottle.

488. \$0.70

489. 10 m/s

490. 14.4 km/h

491. $15\,000 \div 60 = 250$ mL/min

492. \$19/h

493. 5 : 8

494. \$10 and \$35

495. $1\,200\,000$ cm $= 12$ km

496. $5 \times 3.6 = 18$ m^2

497. \$25.50

498. $64 \times 1.125 = \$72$

499. \$1060

500. 45 min

501. 2.83 h

502. 5400 m

503. $\frac{x}{3} = 6$, so $x = 18$.

504. Side $= 7$ m, so perimeter $= 28$ m.

505. 1.8

506. In order: 1, 2, 4, 6, 7, 9. Median $= \dfrac{4+6}{2} = 5$.

507. $\dfrac{2}{6} = \dfrac{1}{3}$

508. 15.4